



Xar Format Specification

An open standard file format for vector graphics on the Web.

Abstract

The Xar file format, previously known as the Flare file format, is an ultra-compact, open, vector graphic format. It is also the native graphics format for Xara X application (and also its predecessors such as CorelXARA).

This document describes the format in detail and provides information for third parties interested in converting to or from this graphics format.

Why another vector graphics format? The Xar file format is not new. It dates back nearly ten years and so it predates more recent formats such as SVG. It is not designed to compete with SVG, but Xar files are *considerably* simpler to understand (the SVG spec is 700 pages) and, more compact (often 10:1). However the primary reason for the existence of the open file format specification is to enable third parties to read and write the Xara X native files.

Background

The Xar vector graphic structure is that of modern vector programs, based on the Adobe Postscript rendering model, but with additional features. The graphic primitives are broadly similar to those available in Postscript, PDF and SVG. However Xara X, and thus the .xar file format, support more advanced graphic primitives and effects, for example a greater range of graduated fill types, graduated transparency, feathered vector shapes (soft edges), soft shadows and more. These enable the talented artist to create highly realistic and more 'painterly' effects with Xara X than is possible with traditional vector graphics solutions. See <http://www.xara.com/gallery/>

The format is designed so that rendering can be started before the entire file is available. The format is extendible, with a degree of forwards and backwards compatibility (this means older version of the software can still read files produced by newer version of the authoring software – they simply ignore the objects or parts they do not understand).

Part of the reason for this compactness is that Xar files are binary, rather than plain text as is SVG (which is XML). But with the recent movement towards a binary XML format, in order to overcome the verbose nature of XML (and SVG), it seems things might come full

circle. On the other hand Xara X offers a plain text variant of the .xar file (file extension .wix) which can help in understanding the structure of .xar files.